with the Linux kernel? What is the difference between a *'noop'* scheduler and a *'deadline'* scheduler?
15. From version 2.4, the Linux kernel supports a unified buffer cache mechanism. Why is a unified buffer cache better than having a separate page cache and a separate buffer cache?

Programming questions

16. You are given a binary search tree and an integer *'k'*. Find out if there exist two nodes of the binary search tree that sum up to *'k'*.
17. You are given *'N'* integers without any duplicates where *'N'* is an even number and a value *'k'*. You are asked to form *'N/2'* distinctive pairs from these *'N'* integers such that the sum of each pair of numbers is divisible by the given *'k'*. Write a program to determine whether such N/2 pairs can be formed and, if so, output the N/2 pairs.
18. Write a program to determine whether a given binary tree is a binary search tree (BST). This is one of those 'Gotcha' questions where most folks tend to use the following check: verify at each node of the BST that it satisfies the following condition —that the value of the left child is less than or equal to the root and the value of the right child is greater than or equal to the value of the root node. Beware that this is not an adequate check. Can you give an example of a binary tree that satisfies this check at each node and still ends up not being a binary search tree?
19. Write an algorithm to determine the k-th to the last element in a singly linked list. If 'k' is greater than the size of the list, the snippet should return NULL.
20. You are given a set of 'N' elements. You are asked to partition the set into two sub-sets such that the sum of elements of each sub-set is the same. Write an algorithm to determine whether a given set can thus be partitioned.
21. You are given an NXN matrix of integers. Each row and each column are sorted. You are asked to find whether a given integer *'k'* exists in the matrix. What is the complexity of your algorithm?
22. You are given a sorted circular array and are asked to find the largest element in the array. Write a program for this.
23. You are given an endless stream of single digit integers. Whenever you see an integer that is repeating for an odd number of times, output the integer value and *'odd'*. Whenever you see an integer that is repeating for an even number of times, output the integer and *'even'*. What is the best time and space complexity algorithm you can come up with?
24. You are given an array A of *'N'* integers without duplicates. You are asked to find a pair of indices *'i'* and *'j'* such that $A[i] < A[j]$ and the difference between *'i'* and *'j'* is the maximum.
25. You are asked to implement a *'Stack'* data structure that supports *'push'* and a modified version of *'pop'* such that *'pop'* removes the most frequently encountered data item in the stack at that point. For instance, given the following sequence of operations:
    (a) Push 3, push 4, push 4, followed by *'pop'* should yield 4
    (b) Push 3, push 4, push 4, push 2, push 3, push 3 followed by *'pop'* should yield 3.

    Note that the user can call *'pop'* at any point after any number of pushes.

My 'must-read book' for this month

This month's 'must-read book' suggestion comes from Gaurav. He recommends the book 'Joel on Software' by Joel Spolsky. Joel hosts the programming blog www.joelonsoftware.com. According to Gaurav, "This book is sub-titled 'Joel On Software And on Diverse and Occasionally Related Matters That Will Prove of Interest to Software Developers, Designers, and Managers, and to Those Who, Whether by Good Fortune or Ill Luck, Work with Them in Some Capacity' and essentially covers topics right from coding to product management. Joel Spolsky is a legend in the programming world and his caustic humor and deep insights come through strongly in this collection of essays about software." Thank you Gaurav, for your suggestion.

If you have a favourite programming book/article that you think is a must-read for every programmer, please do send me a note with the book's name, and a short write-up on why you think it is useful so that I can mention it in this column. It would help many readers who want to improve their software skills.

If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at *sandyasm_AT_yahoo_DOT_com*. Till we meet again next month, happy programming and here's wishing you the very best! END

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The author is an expert in systems software and is currently working with Hewlett Packard India Ltd. Her interests include compilers, multi-core and storage systems. If you are preparing for systems software interviews, you may find it useful to visit Sandya's LinkedIn group 'Computer Science Interview Training India' at *http://www.linkedin.com/groups?home=HYPERLINK "http://www.linkedin.com/groups?home=&gid=2339182"&HYPERLINK "http://www.linkedin.com/groups?home=&gid=2339182"gid=2339182*